Research on the Prediction Model of University Library Borrowing Behavior Based on Big Data and Machine Learning

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***Abstract*—The use of machine learning algorithms for deep data mining is aimed at improving the efficiency of public library borrowing recommendations. According to the characteristics of users' reading behavior, a heuristic recommendation method which is more suitable for users' use behavior is designed. Using RBF network to predict the information of book borrowing, a method to predict the number of book borrowing is constructed. The method is compared with other prediction methods. The results show that this method can estimate the quantity of collection accurately and reduce the error rate of estimation. The prediction method proposed in this paper is effective. The recommendation algorithm model shows high application efficiency in simulation test and trial run test, and the subjective evaluation results are given by members who have used the new system. Through the analysis of literature retrieval results, empirical evidence confirms the viability of the search process paradigm predicated on heuristic discovery presented within this document.**

Keywords—Borrowing behavior, machine learning, data mining technology, ant colony algorithm, library management, RBF neural network

I. Introduction

In recent years, due to the preference of readers in using audio books and electronic books, the borrowing market of paper books in public libraries has been greatly affected. However, since the public library is a non-profit organization, it needs social benefits rather than economic benefits. The cultural capacity and knowledge dissemination ability of paper books should be greatly enhanced. Therefore, at present, public libraries in various places carry out digital transformation of traditional paper books [1]. This project intends to start with the big data analysis of user behavior, establish a heuristic borrowing process based on user preferences, and provide users with books that meet their needs and achieve predetermined services.

II. A Fully Electronic Big Data System for Public Libraries

The data resources of public libraries include three aspects: 1) Digital book big data: The text content in paper books is converted into electronic text in various languages through modular data processing means such as laser scanning, text recognition, and machine translation. Some libraries can also convert paper books into sound through modular processing means such as machine reading, so as to construct a complete set of audio book data. Thus, the online and offline lending of electronic books and paper books can be achieved. 2) Book lending card big data: including title, author, publisher, publication time, print and print number, open edition, page number, document identification code, Chinese map classification number, multi-language version abstract, keywords, etc., constitute the basic elements of traditional book lending cards, they can form the logic of query, so as to achieve the purpose of batch query. In the context of big data, the word frequency feature code and style identification code of books will also be included in the big data of book borrowing cards, so as to achieve the logic of the entire query. 3) Member big data: Member's name, age, gender, ethnicity, unit, occupation, online borrowing records, physical book borrowing records, etc., constitute the big data of members [2]. This paper analyzes the big data of members, and then constructs the heuristic recommendation process of members' information. the logical correlation between the above three aspects of big Data is shown in Figure 1 (the picture is quoted in Legal Protection of Artificial Intelligence Data and Algorithms from the Perspective of Internet of Things Resource Sharing).

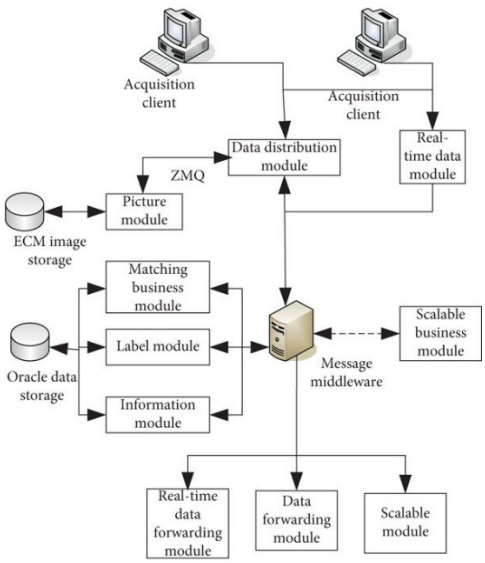


Fig. 1. Basic logical architecture diagram of the heuristic borrowing process model

Finally, the system categorizes the book proposals for borrowers into four categories based on priority: 1) I borrowed books with the same keywords or similar titles

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and contents in historical books for many times with members. From the perspective of audience psychology, these books are purely rational, and members are most willing to be accepted by members, but it cannot be ruled out that they want to change the category of borrowing to experience new books. 2) If the user has borrowed the same author or author's history books for many times, the author is related to the two authors have published similar works for many times, or there is a relationship between the two authors, such as a user has borrowed Jia Pinga's works for many times, then the platform can recommend Jia Jianqian's works to him. 3) Recommend books that are in line with readers' work content and nature. At present, professional experience and counseling books account for a large proportion of printed books. If the readers are civil servants or corporate executives, recommend some books on management and psychology; if the readers are engineers, technicians or laborers, then recommend some technical coaching books. 4) It is recommended that users choose books related to this keyword, such as title, keyword, abstract, content word frequency characteristics, etc.

III. Related Technologies and Workflow of Electronic Physical Books

In the current scientific and technological environment, digital work can only complete the conversion of some text from paper books to electronic text, as well as subsequent operations (such as machine translation, vocabulary extraction, machine reading, etc.), and the digital process of paper books is the key support for its implementation [3]. Compared with simply using the library card of electronic books to recommend books, the digitized books can depict the characteristics of books more completely and meet the needs of computer-aided book recommendation algorithm in cloud computing environment. The underlying logical structure of this process is shown in Figure 2 (image cited in Using Smart Meters Data for Energy Management Operations and Power Quality Monitoring in a Microgrid).

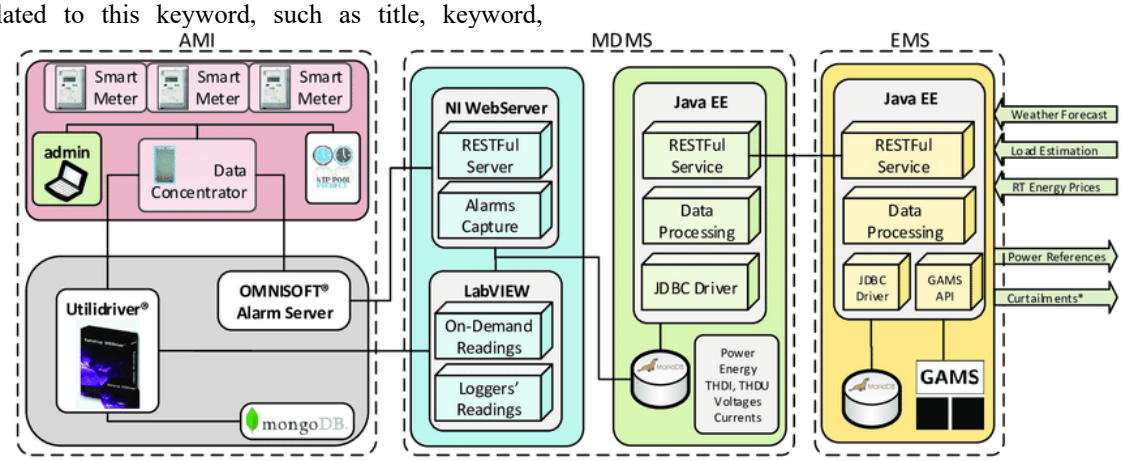


Fig. 2. Logical architecture diagram of electronic work flow of physical books

This project intends to use Hanwang laser scanning system, Baidu translation system for machine translation, and IFLYtek machine reading system for word frequency analysis, neural network analysis and other functions [4]. The system realizes the digital information of the book content which can be recognized by machine learning function through the style feature code and word frequency feature code. The original language version after text recognition and the multi-language version after machine translation are also used in the direct retrieval process.

IV. Steps of Extracting and Using Frequency Characteristic Code and Style Code

The basic structure of the system is the front and back overlap of two 4-byte floating point type variables. Its initial Generation algorithm framework is shown in Figure 3 (image cited in Automatic Code Generation for the Orchestration of Web Services with Reo).

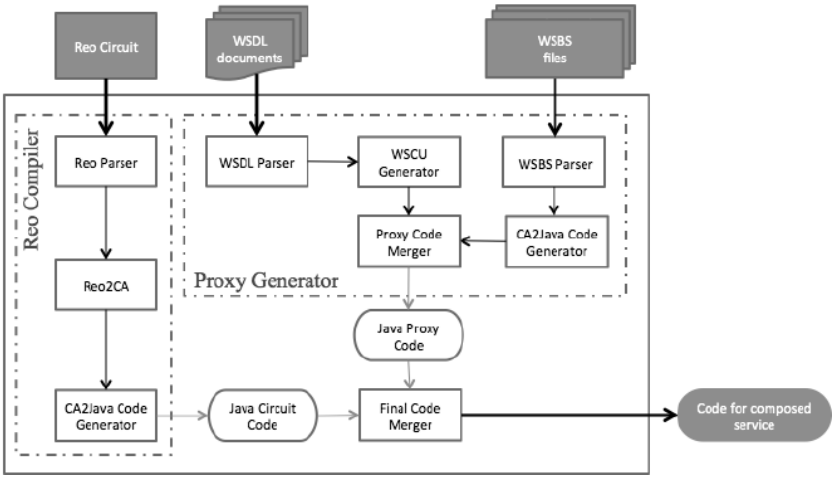


Fig. 3. Logical architecture diagram of feature code generation algorithm

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Figure 3 uses two convolutional neural networks with the same structure and input data to generate two independent data columns. According to different needs, the two network models are converged in their own ways, and then the two convolutional neural networks are introduced into the encoded high 32 bit and low 32-bit deep convolutional neural networks [5]. The neural network generates a code of up to 32 bits, and on the output of the other side network, 1 auxiliary data is extracted. It's finally overwritten by a 64-bit code. With the exception of some differences in the input data, the logical framework generated by the stylistic code is identical. The input data to generate the lexical feature code is the lexical order data obtained by the frequency extraction algorithm of the electronic book, and the input data to generate the book-style code is the original data of the book. The two feature codes are extracted, compared and fused with the members' reading habit codes. The machine learning algorithm determines the similarity between the members' reading habit codes and the two feature codes of the book, and puts forward a recommended order. When the user uses the book, the member feature codes of the user will be updated and rewritten according to the two feature codes of the book. The logical structure of this process is shown in Figure 4 (the picture is quoted in Entropy 2021, 23(5), 529).

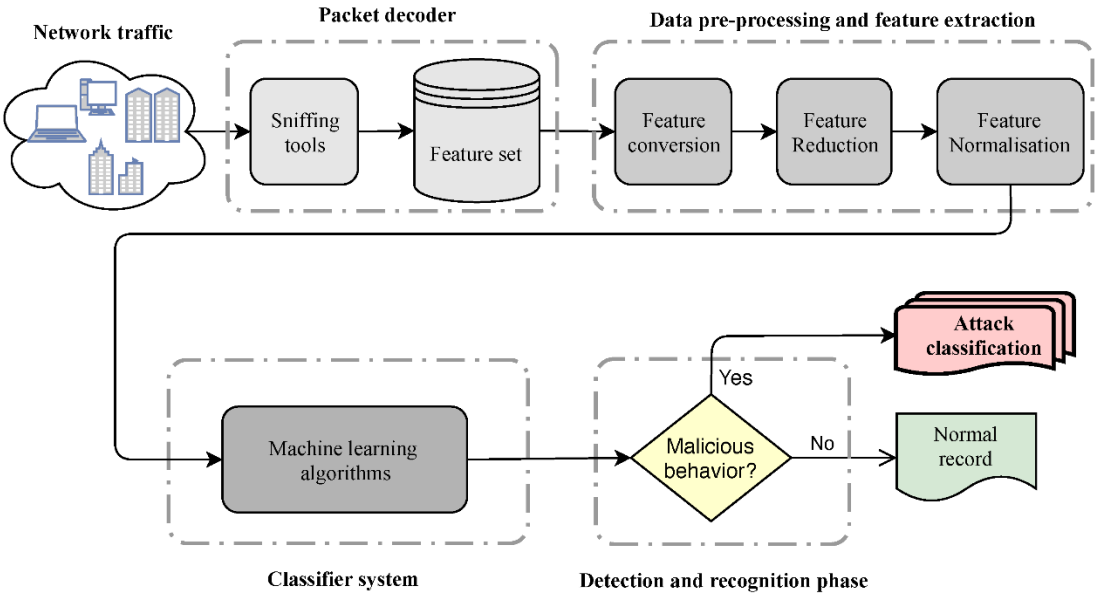


Fig. 4. Logical architecture diagram of feature code application and transformation mechanism

The binarized neural network in Figure 4 gets a decision [0,1]. Through the training of the neural network, a small number of books tend to be the recommended choice in the direction of 000, while other books tend to be 0.000 and are therefore filtered out. When the member is lending, it is input into the convolutional neural network together with the written feature code and the member feature code to generate the fused new member feature code, and then the membership card digital information is reassigned to the next decision procedure [6]. This paper presents a new member coding method based on new member coding, which takes the actual time of borrowing books, reading time of electronic books and audio books as the new member coding design method.

Aiming at the first of the four suggestions mentioned above, this project innovatively designs a machine learning-based book recommendation method. This method completely avoids the two defects of the existing method: 1) Due to the limitation of the basic data of the card, no matter the keywords, titles, abstracts, etc., cannot fully reflect the types of books, and using this method, when the number of nodes in the network and the size of the hidden layer reach a critical value, the reader's next borrowing behavior can be more accurately determined. 2) Combining the traditional algorithm with the innovative algorithm, taking the algorithm as a preferred recommendation category and retaining the recommendation results of the traditional recommendation algorithm in other recommendation sectors can make the two forms beneficial complementarity.

V. Book Borrowing Rate Estimation Model Based on Data Mining

A. Radial basis function neural network

The RBF neural network transforms the input information into a higher dimension through the hidden layer and turns it into a decomposing problem[7]. The radial basis function in the RBF neural network is as follows:

$$f_i = \exp\left[-\frac{\| x-\varepsilon_i \|}{2\delta_i^2}\right] \tag{1}$$

Where δ_i and ε_i represent the RBF neural network and the node center of the i hidden unit of the RBF neural network, respectively. Adjusting the neuron weights and decision boundaries significantly impacts the performance of Radial Basis Function (RBF) neural networks, necessitating the identification of ideal parameters for achieving peak precision in predictions. Inspired by natural processes, the artificial ant colony optimization technique emerges as a novel approach. Its application to scheduling and inventory management issues has yielded encouraging outcomes. By integrating this ant colony optimization into the RBF network parameter tuning process, we enhance the network's estimation accuracy.

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B. Ant colony algorithm

1) Initialize ant colony location and pheromone

The size of ant colony is represented by N, the location of ant colony is initialized, and the optimal pheromone of a group of ants can be obtained according to the parameter value of RBF network.

$$\Delta h(i) = \exp(-g'(u_i)) \tag{2}$$

Pheromone concentration tends to 0 when it is at $g(u_i) \geq 0$ and close to infinity. In order to improve the estimation accuracy, the fitness value $g(u_i)$ of ants needs to be corrected. The modification formula for the fitness value of ants is as follows:

$$g'(u_i) = \begin{cases} g(u_i)/avg, & avg > avg_0 \\ g(u_i), & other \end{cases} \tag{3}$$

$g(u_i)$ represents the fitness value before correction; $g'(u_i)$ represents the modified fitness value; avg represents the average fitness value before correction.

2) Path Selection Rules

After one search, ants select the next search path according to the selection rule, randomly select ants with k in the ant colony, extract the individual with the largest pheromone concentration from the selected ants and set it as the transfer target U_{obj}. The transfer target formula is as follows:

$$U_{obj} = \begin{cases} U_j, & h(U_i) < \max(h(U_j)) \\ U_{best}, & other \end{cases} \tag{4}$$

Where U_{best} is the optimal solution obtained in the last iteration. As the sex hormone increases, so does the attraction to the ants, so the likelihood of obtaining the best solution is higher, and according to formula (5) the ants will converge to a target location.

$$U_i = (1-\eta)U_i + \eta U_{obj} \tag{5}$$

At the end of the last step, the best ant from the most recent iteration was obtained, and then carefully searched for it in the adjacent areas, resulting in the following global optimal solution.

$$U_{best} = \begin{cases} U_i', g(U_i') < g(U_{best}) \\ U_{best}, \quad other \end{cases} \tag{6}$$

VI. Confirmatory Experiments

Taking a university library as a research sample, Visual C++6.0 is used to develop the model, and the correctness of the method is verified by this method [8]. Through this set of university loan management platform, the borrowing situation between January 1 and December 31, 2022 was obtained, and the name, age, gender, major and the borrowed books were statistically analyzed. Use the collected raw data to estimate the total annual library loans of the university in 2023 (Figure 5).

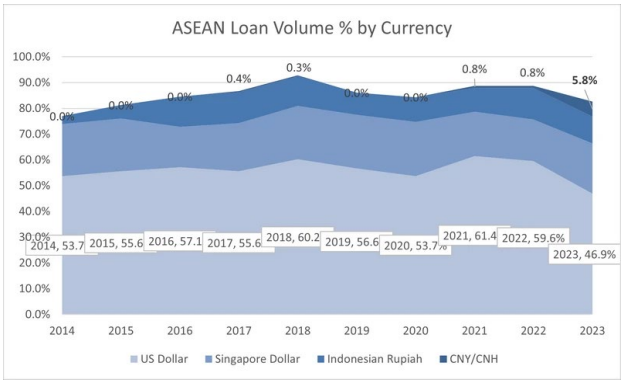


Fig. 5. Estimated library loan volume in 2023

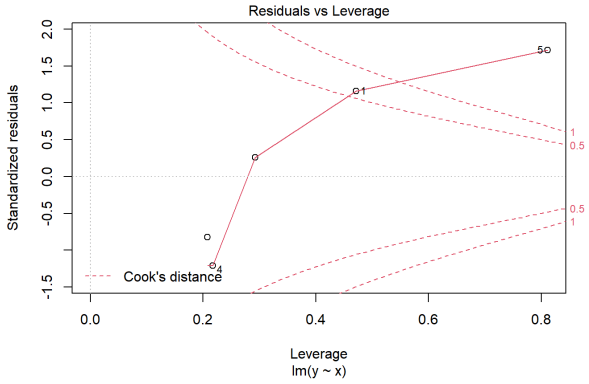


Fig. 6. Error in estimation of library loan volume

This model is used to estimate the utilization rate of schools in 2023 and universities in 2022 to test the accuracy of the prediction in this paper [9]. It can be seen from the experimental data in Figure 6 that this model can well estimate the amount of loan in each month in the next two months, and the deviation of its estimate is less than 2.5%, which indicates that this model is a good evaluation model. In order to test the number of books borrowed by this model in the library, this paper uses this model to forecast various books in the school library in 2023. The estimated data are

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shown in Table I. From the data in Table I, this paper can see that this model can be used to accurately estimate various types of books in the school library, and estimate the borrowing rate of various types of books, and the predicted error is within 2.5%.

TABLE I. ESTIMATES OF LOAN VOLUME FOR DIFFERENT CATEGORIES OF BOOKS

Book category	Lending error /%
Language	1.17
computer	0.48
literature	1.03
Chemistry	0.23
mathematics	1.22
economy	0.20
industry	0.24
Philosophy, Religion	0.20
Politics and law	0.70
Physical education	2.01
education	0.48
culture	2.51
Environmental science	0.41
medicine	0.09
Social science	1.29
Military affairs	0.11
astronomy	0.36
Earth science	1.42
synthesis	0.92
Natural science	0.86
bioscience	0.57
transportation	0.13
Agricultural science	0.25
unit	0.46
secure	0.24

Among the library cards of university libraries, while the proportion of teachers and graduate students is above the average, one is 25%, the other is 14%, and the other is preparatory students [10]. This is mainly the number of students is relatively small, so the data estimated by the model in this paper is consistent with the actual amount of borrowing by the university. This method can be used to accurately estimate the borrowing situation of various types of readers, and has high practical value.

The paper estimated the number of undergraduate loans of each gender in each grade in 2023, and compared it with the actual borrowing rate of each grade, male and female in 2023, and drew the following conclusions. From the experimental data in Table II, it can be seen that this method can be used to accurately estimate the book borrowing status of students in different grades, and the estimation error is very small [11]. As can be seen from Table II, the number of books borrowed by grade 2019 and 2023 is significantly lower than that of grade 2020, 2021 and 2022. Since students of grade 2019 only study in the first half of 2023, and students of Grade 2023 will enter the school after the semester starts in September, the school time of the above two grades in 2023 will be shorter. Therefore, the loan volume is significantly lower than that of grade 2020, 2021 and 2022; The rate of female students borrowing books is significantly higher than that of male students, indicating that female students invest more in their studies than male students, which is consistent with the reality. The model estimates the number of books borrowed by male and female students in each grade.

TABLE II. ESTIMATES OF THE AMOUNT OF UNDERGRADUATE LOANS BY GENDER IN EACH GRADE

Grade	Male			Female		
	Estimated volume of loans/copies	Actual number of loans/copies	Estimated error /%	Estimated volume of loans/copies	Actual number of loans/copies	Estimated error /%
2019	20988	21305	1.55	27660	27954	1.09
2020	48804	48985	0.39	55467	54535	1.78
2021	52528	52408	0.24	66961	65818	1.81
2022	50647	50899	0.51	55890	56652	1.41
2023	20338	20631	1.48	22244	22008	1.11

VII. CONCLUSION

This paper proposes a library borrowing prediction model based on data mining technology, which solves the nonlinear high-dimensional problem of library borrowing effectively and improves the accuracy of prediction. The accuracy of the model is verified by testing the loan estimation of a university library. This model can not only predict the total borrowing rate of the library, but also predict the borrowing rate of different types of books, different types of readers and different grades of students. The innovation of this study lies in the application of complex data mining techniques to the prediction of library borrowing behavior. The traditional library borrowing prediction mainly relies on statistical analysis or simple mathematical model, which is difficult to deal with a large number of nonlinear characteristics. With the coming of the big data era, the amount of data faced by libraries will continue to increase, and the application prospect of data mining technology in library management is very broad. In the future, the model can further integrate data from more dimensions to improve the comprehensiveness and foresight of predictions. At the same time, the model can also be extended to other types of libraries, such as public libraries, professional libraries, etc., and can even be applied to the demand prediction of other service fields, such as cinema box office prediction and tourist flow prediction. To sum up, the library borrowing prediction model proposed in this paper provides a powerful tool for the scientific management and service of libraries by using data mining technology. This model can not only improve the accuracy of library borrowing prediction, but also help managers better understand the borrowing behavior, so as to optimize the allocation of resources and improve the quality of library service. With the continuous progress of technology and the increasing abundance of data resources, this model is expected to play a greater role in the future and promote the continuous development of library.

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